

Notes – WIP

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1 Introduction

Ez a dokumentum azt hivatott összegezni, amit a kollektív kónikus kereszteződési pontokkal kapcsolatos topologikus modellünkkel eddig elértünk.

Az alapelrendezése a rendszerünknek, hogy cavity-ben, három azonos kétatomos molekulát helyezünk el egy derékszögű koordináta rendszerben, ahol a molekulák tengelyei párhuzamosak az x , y és z tengelyekkel. Ez az elrendezés alapvetően megteremtheti a CI-k kialakulását a rendszerben.

2 Model Potentials

Our system is described by two potential energy surfaces:

$$V_x(x) = a_{V_x} \cdot x^2 \tag{1}$$

$$V_a(x) = a_{V_a} \cdot (x - x_{\text{shift}})^2 + c_0 \tag{2}$$

Where:

- a_{V_x} and a_{V_a} are scaling parameters
- x_{shift} is a shift parameter for the second potential
- c_0 is a constant energy offset

3 Finding Conical Intersection Points

A conical intersection occurs when the two potential energy surfaces meet at a point. For our model, this happens when:

$$V_a(x) - V_x(x) = 0 \tag{3}$$

3.1 Mathematical Derivation

Starting with the difference between the potentials:

$$V_a(x) - V_x(x) = a_{V_a} \cdot (x - x_{\text{shift}})^2 + c_0 - a_{V_x} \cdot x^2 \quad (4)$$

$$= a_{V_a} \cdot x^2 - 2 \cdot a_{V_a} \cdot x_{\text{shift}} \cdot x + a_{V_a} \cdot x_{\text{shift}}^2 + c_0 - a_{V_x} \cdot x^2 \quad (5)$$

$$= (a_{V_a} - a_{V_x}) \cdot x^2 - 2 \cdot a_{V_a} \cdot x_{\text{shift}} \cdot x + c_1 \quad (6)$$

Where $c_1 = a_{V_a} \cdot x_{\text{shift}}^2 + c_0$.

We can rewrite this in the form of a perfect square plus a constant:

$$V_a(x) - V_x(x) = (a_{V_a} - a_{V_x}) \cdot (x - x_{\text{prime}})^2 + c'_1 \quad (7)$$

Where x_{prime} is the point where the quadratic term has its minimum:

$$x_{\text{prime}} = \frac{a_{V_a}}{a_{V_a} - a_{V_x}} \cdot x_{\text{shift}} \quad (8)$$

4 Geometric Configuration of CI Points

For a 3D system with coordinates (r_0, r_1, r_2) , we need to find points where the potentials are equal. This leads to a specific geometric arrangement.

4.1 Key Insight

If we have a point $R_0 = (r_0, r_1, r_2)$ where two of the coordinates give equal potential differences, then we can find a CI point. Specifically, if:

$$(V_a - V_x)(r_1) = (V_a - V_x)(r_2) \quad (9)$$

Then we can express this in terms of distances from x_{prime} :

$$r_1 = x_{\text{prime}} - \delta \quad (10)$$

$$r_2 = x_{\text{prime}} + \delta \quad (11)$$

Where δ is some displacement.

4.2 Relationship Between Coordinates

Given that only two coordinates can be equal at the same time to create a CI point, we can derive the following relationships:

If we set $r_0 = x_{\text{prime}} - \alpha$ for some value α , then:

$$r_2 - r_0 = -2 \cdot (r_1 - r_0) \quad (12)$$

This leads to:

$$2(r_0 - r_1) = r_2 - r_0 \quad (13)$$

$$2r_0 - 2x_{\text{prime}} + 2\delta = x_{\text{prime}} + \delta - r_0 \quad (14)$$

$$3r_0 = 3x_{\text{prime}} - 3\delta \quad (15)$$

$$\delta = 3(x_{\text{prime}} - r_0) \quad (16)$$

4.3 Expressions for r_1 and r_2

Using the value of δ , we can express r_1 and r_2 in terms of r_0 and x_{prime} :

$$r_1 = x_{\text{prime}} - \delta \quad (17)$$

$$= x_{\text{prime}} - 3(x_{\text{prime}} - r_0) \quad (18)$$

$$= 3r_0 - 2x_{\text{prime}} \quad (19)$$

$$= r_0 - 2(x_{\text{prime}} - r_0) \quad (20)$$

Similarly:

$$r_2 = x_{\text{prime}} + \delta \quad (21)$$

$$= x_{\text{prime}} + 3(x_{\text{prime}} - r_0) \quad (22)$$

$$= 4x_{\text{prime}} - 3r_0 \quad (23)$$

$$= r_0 + 4(x_{\text{prime}} - r_0) \quad (24)$$

5 Distance to CI Points

The distance from R_0 to the CI point can be calculated as:

$$d_{\text{CI}} = (r_2 - r_0) \cdot \frac{\sqrt{6}}{2} \quad (25)$$

$$= 4(x_{\text{prime}} - r_0) \cdot \frac{\sqrt{6}}{2} \quad (26)$$

$$= 2\sqrt{6} \cdot (x_{\text{prime}} - r_0) \quad (27)$$

6 Implementation in Our Code

In our code, we implement this by setting:

$$r_0 = x_{\text{prime}} \quad (28)$$

$$x = 2(x_{\text{prime}} - r_0) = 0 \quad (29)$$

For the CI point at index n_{CI} , we set:

$$R_0[i] = \begin{cases} r_0 + x + x & \text{if } i = n_{\text{CI}} \\ r_0 - x & \text{otherwise} \end{cases} \quad (30)$$

With our parameter values, this gives $R_0 = [0.433, 0.433, 0.433]$ when $n_{\text{CI}} = 3$ (i.e., no specific CI point selected). We should run both a θ^* - and a d^* -scan on this specific point.

6.1 Arrowhead Hamiltonian

Since the cavity excites collectively, the system is described by a 4×4 arrowhead Hamiltonian matrix with the following structure:

$$H(\mathbf{R}_\theta) = \begin{pmatrix} \hbar\omega + \sum_i V_x(\mathbf{R}_i) & t_{01} & t_{02} & t_{03} \\ t_{01} & V_e(\mathbf{R}_0) & 0 & 0 \\ t_{02} & 0 & V_e(\mathbf{R}_1) & 0 \\ t_{03} & 0 & 0 & V_e(\mathbf{R}_2) \end{pmatrix} \quad (31)$$

where:

- $\hbar\omega$ is the energy quantum of the system
- $V_x(\mathbf{R}_i) = a_{Vx}|\mathbf{R}_i|^2$ is the potential energy function
- $V_e(\mathbf{R}_i) = \sum_j V_x(\mathbf{R}_j) + V_a(\mathbf{R}_i) - V_x(\mathbf{R}_i)$ is the effective potential
- $V_a(\mathbf{R}_i) = a_{Va}(|\mathbf{R}_i - x_{\text{shift}}|^2 + c)$ is the additional potential
- t_{0i} are the transition dipole moments between states

The parameter vector $\mathbf{R}_\theta$ traces a perfect circle orthogonal to the $x = y = z$ line in 3D space, parameterized by angle θ .

6.2 Berry Connection and Berry Phase

The Berry connection $\mathbf{A}_n(\mathbf{R})$ for the n -th eigenstate is defined as:

$$\mathbf{A}_n(\mathbf{R}) = i \langle n(\mathbf{R}) | \nabla_{\mathbf{R}} | n(\mathbf{R}) \rangle \quad (32)$$

The Berry phase γ_n is the integral of the Berry connection around a closed loop C in parameter space:

$$\gamma_n = \oint_C \mathbf{A}_n(\mathbf{R}) \cdot d\mathbf{R} \quad (33)$$

6.3 Off-Diagonal Elements

The off-diagonal elements $\tau_{nm}(\theta)$ for $n \neq m$ provide information about the coupling between different eigenstates. These elements can be used to analyze transitions between states and to identify regions in parameter space where eigenstates are strongly coupled.

7 The Tau Matrix Method

We call this the "tau matrix method" because the key matrix is the tau matrix in the codebase. However, this matrix is actually the matrix of the Non-adiabatic coupling terms.

In this method, we first compute τ matrix, then use it to compute the Berry phase.

7.1 Definition of the Tau Matrix

This method is a numerical approach for computing the Berry phase. The key element is the tau matrix, whose elements are defined as:

$$\tau_{nm}(\theta) = \langle \psi_m(\theta) | \frac{\partial}{\partial \theta} | \psi_n(\theta) \rangle \quad (34)$$

where $|\psi_n(\theta)\rangle$ is the n -th eigenstate of the Hamiltonian $H(\mathbf{R}_\theta)$.

8 Numerical Results and Analysis

8.1 Berry Phase Matrix Structure

The numerical results show that the Berry phase matrix γ_{nm} has a specific structure with the following features:

- The diagonal elements γ_{nn} are typically small (order 10^{-3} to 10^{-4}), indicating minimal geometric phase accumulation for individual states.
- Large off-diagonal elements (order 10^0) appear in specific positions, particularly between states 1-2, 2-1, 1-3, and 3-1.
- The values close to $\pm 2\pi$ in the [1,2] and [2,1] positions indicate complete phase winding, which is a signature of non-trivial topology.

8.2 Physical Interpretation

The observed Berry phase structure has important physical implications:

- The system exhibits non-trivial geometric phase accumulation primarily in the subspace spanned by states 1 and 2.
- The antisymmetric nature of the off-diagonal elements ($\gamma_{12} \approx -\gamma_{21}$) is consistent with the expected behavior of a quantum system with time-reversal symmetry.
- The 2π phase difference indicates that the system's parameter space trajectory encloses a topological singularity.